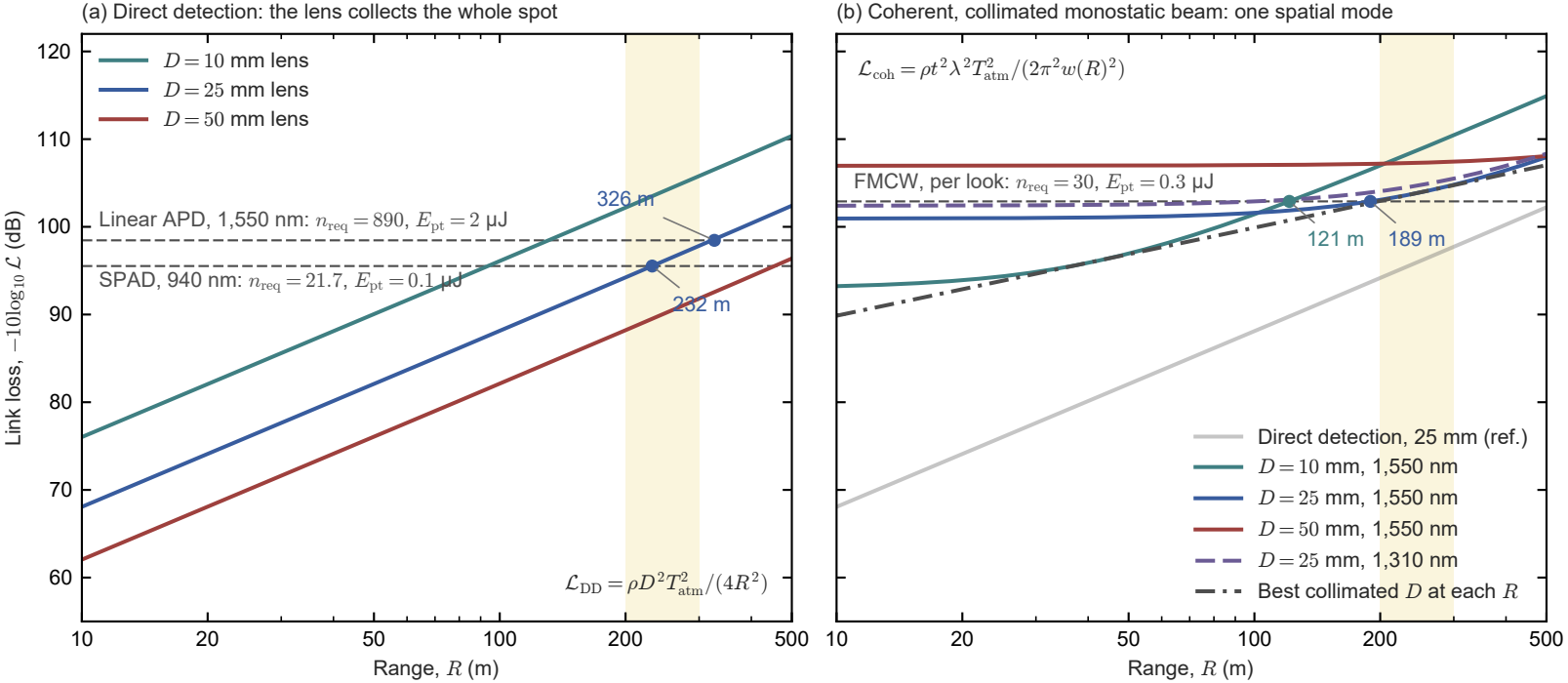
(a) Direct detection: the lens collects the whole spot
$D = 10$ mm lens
$D = 25$ mm lens
$D = 50$ mm lens
Linear APD, 1,550 nm: $n_{\mathrm{req}} = 890$, $E_{\mathrm{pt}} = 2$ μJ
SPAD, 940 nm: $n_{\mathrm{req}} = 21.7$, $E_{\mathrm{pt}} = 0.1$ μJ
326 m
232 m
$\mathcal{L}_{\mathrm{DD}} = \rho D^2 T_{\mathrm{atm}}^2/(4R^2)$
Link loss, $-10\log_{10}\mathcal{L}$ (dB)
Range, R (m)
(b) Coherent, collimated monostatic beam: one spatial mode
$\mathcal{L}_{\mathrm{coh}} = \rho t^2 \lambda^2 T_{\mathrm{atm}}^2/(2\pi^2 w(R)^2)$
FMCW, per look: $n_{\mathrm{req}} = 30$, $E_{\mathrm{pt}} = 0.3$ μJ
121 m
189 m
Direct detection, 25 mm (ref.)
$D = 10$ mm, 1,550 nm
$D = 25$ mm, 1,550 nm
$D = 50$ mm, 1,550 nm
$D = 25$ mm, 1,310 nm
Best collimated D at each R
Range, R (m)